2.1 Percentage increase

The price of a designer T-shirt was $85.
The shop increases the price by 20%.

To calculate the new price, we increase $85 by 20%.

When a quantity is increased by a given percentage, it is made larger. The percentage amount is **added** to the original quantity.

EXAMPLE

Increase $85 by 20%.

Method 1:	*Method 2:*
20% of $\$85 = \frac{20}{100} \times 85 = \$17$ The increased amount is $\$85 + \$17 = \mathbf{\$102}$	The new amount is $100\% + 20\% = 120\%$ of the original 120% of $\$85 = \frac{120}{100} \times 85 = \mathbf{\$102}$

1 Increase each amount by 20%.

a $60

b $150

c $230

d 750 mL

e 80 kg

f 4.8 metres

2 Complete these percentage increase calculations.

a Increase $50 by 10%.

b Increase $200 by 12%.

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2.1 Percentage increase

CONTINUED

c Increase $42 by 30%.

d Increase $168 by 25%.

e Increase $690 by 15%.

f Increase $28.50 by 10%.

g Increase 80 km by 40%.

h Increase 60 tonnes by 5%.

i Increase 3.2 kg by 15%.

j Increase $120 by 12.5%.

Answer the following questions in your workbook.

3 A pair of sports shoes is priced at $130.
What is the new price if the shop increases all prices by 15%?

4 A satchel cost $72 to make.
Calculate the selling price if the cost is marked up by 40%.

5 Jamal's current weekly pay is $740.
His pay is increased by 3%.
What is his new weekly pay?

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2.2 Percentage decrease

A skateboard is priced at $230.
During a sale, the price is reduced by 10%.

To calculate the sale price, we **decrease $230 by 10%.**

When a quantity is decreased by a given percentage, it is made smaller. The percentage amount is **subtracted** from the original quantity.

EXAMPLE

Decrease $230 by 10%.

Method 1:

10% of \$230 = $\frac{10}{100} \times 230$
= \$23

The decreased amount is \$230 – \$23
= \$207

Method 2:

The new amount is 100% – 10%
= 90% of the original

90% of \$230 = $\frac{90}{100} \times 230$
= \$207

1 Decrease each amount by 10%.

a $80 **b** $130 **c** $65

d 200 km **e** 450 litres **f** 9.8 kg

2 Complete these percentage decrease calculations.

a Decrease $200 by 12%. **b** Decrease $120 by 15%.

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2.2 Percentage decrease

CONTINUED

c Decrease $340 by 25%.

d Decrease $1 570 by 40%.

e Decrease $645 by 20%.

f Decrease $18 by 5%.

g Decrease 120 metres by 45%.

h Decrease 160 kg by 20%.

i Decrease 70 litres by 15%.

j Decrease 200 km by 12.5%.

Answer the following questions in your workbook.

3 The population of a town was 18 400. It has decreased by 8%.
What is the current population of this town?

4 A white-water rafting weekend is priced at $1 850.
During a sale, the price is reduced by 30%.
What is the sale price?

5 Katie's weekly wage is $835. This is decreased by 25% when income tax is deducted.
How much remains?

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